# MediChain

The emergency summary that travels with the patient.
Every access to it is visible to them.

A South African health ID, shared care record and emergency-access platform for patients, clinicians and responders: live on AWS in Cape Town.

https://13-247-143-188.sslip.io/patient/ · https://13-247-143-188.sslip.io/doctor/

2nd place, Rust Africa Hackathon 2026

# When records go missing, care slows down

## Limpopo, 40 hospitals

Missing records meant patients "waiting longer to be treated" and some "treated without medical history".

Marutha & Ngoepe, SAJIM 2017

## Across South Africa

"Numerous South African public health facilities fail to meet their record-keeping obligations as required by law."

Kolawole, AJIC 2024

## In emergencies

At Khayelitsha Hospital, un-scanned paper folders left months of emergency-centre records unavailable electronically.

Ohuabunwa et al., AfJEM 2016

National policy asks for person-centred digital health and a common patient identifier (Digital Health Strategy 2019–2024). Patients rarely see who reads their record.

Studies describe the problem; they are not MediChain outcomes. The Khayelitsha study covers one hospital.

# One evening, two endings

Thabo, 55: diabetes, high blood pressure, allergic to aspirin. At 18:40 he collapses at a taxi rank in Tembisa.

| Time | Today, with paper folders | With MediChain |
|---|---|---|
| 18:52 ambulance | Thabo cannot speak; nobody knows his conditions or allergies | The paramedic scans the QR on his lock screen: allergies, medicines, next of kin |
| 18:58 treatment | Chest-pain protocols often include aspirin; nobody can answer | The aspirin allergy is first on screen; a 15-minute grant for Thabo only |
| 19:30 handover | His folder is at a clinic 20 km away | The hospital doctor opens the same record |
| Next morning | He cannot know who read his information | His access history lists the paramedic and the doctor, with reasons |

An illustrative scenario, not a measured outcome.

# One record, three promises

## Emergency in seconds

Scan the patient's QR with any camera, tap their NFC card on Android, or type the card number. Blood type, allergies, medicines and contacts appear on an approved device, under a time-limited grant.

## Patients see every access

Who opened the record, when and why, including emergency access. Patients grant and withdraw staff access and sign or refuse consent.

## The care team on one record

Doctor, nurse, pharmacist, lab technician and paramedic work from the same notes, orders, prescriptions, results and handovers, and message the patient.

No connection? The QR carries a printed summary that still shows offline, marked unverified.

# How an emergency read works

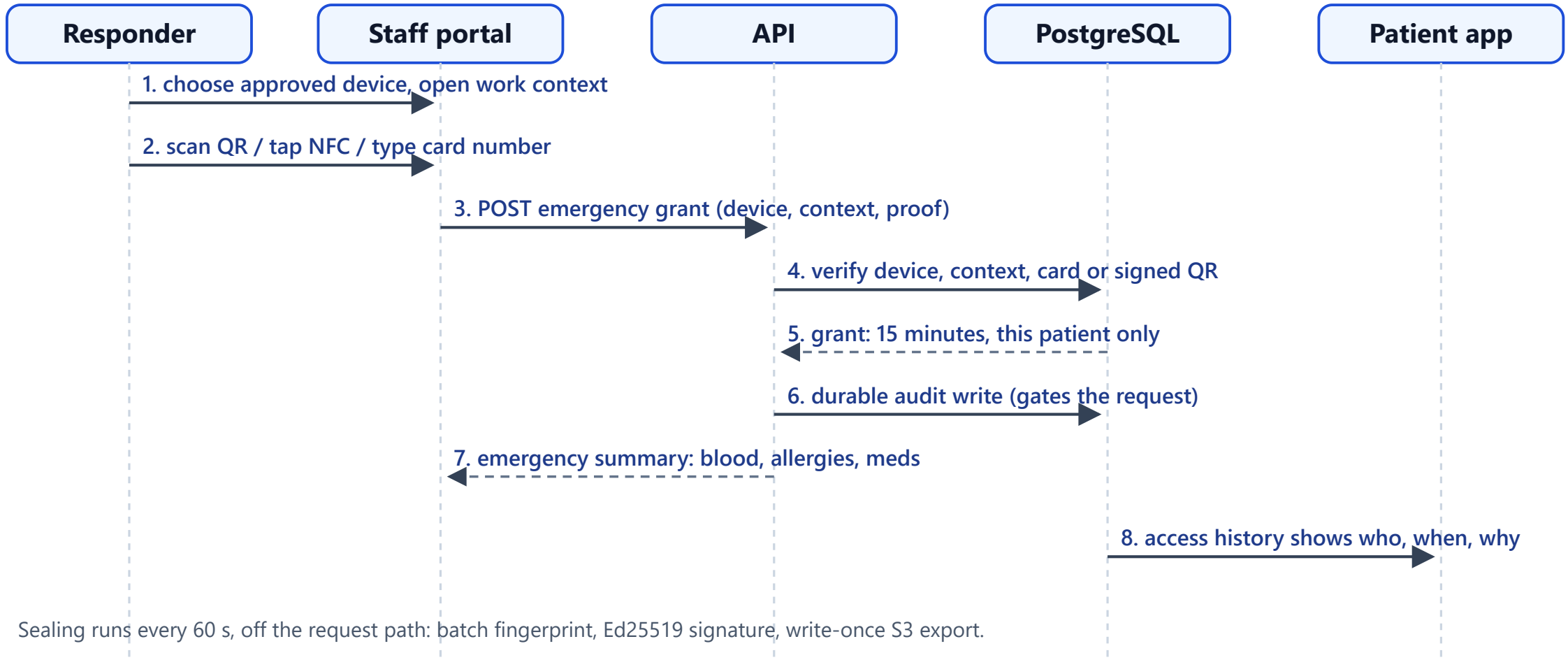
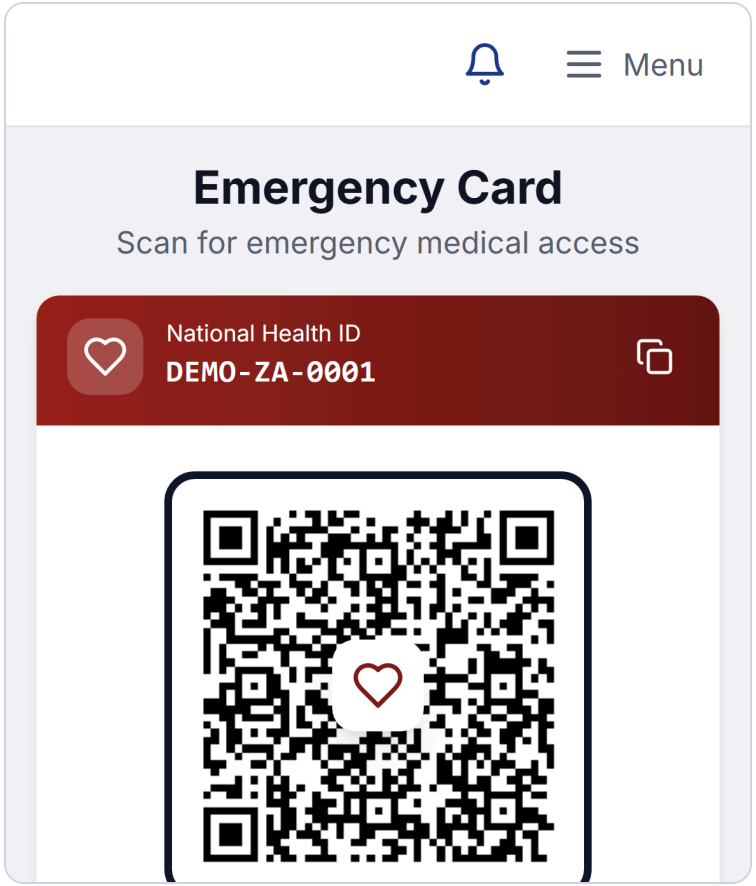
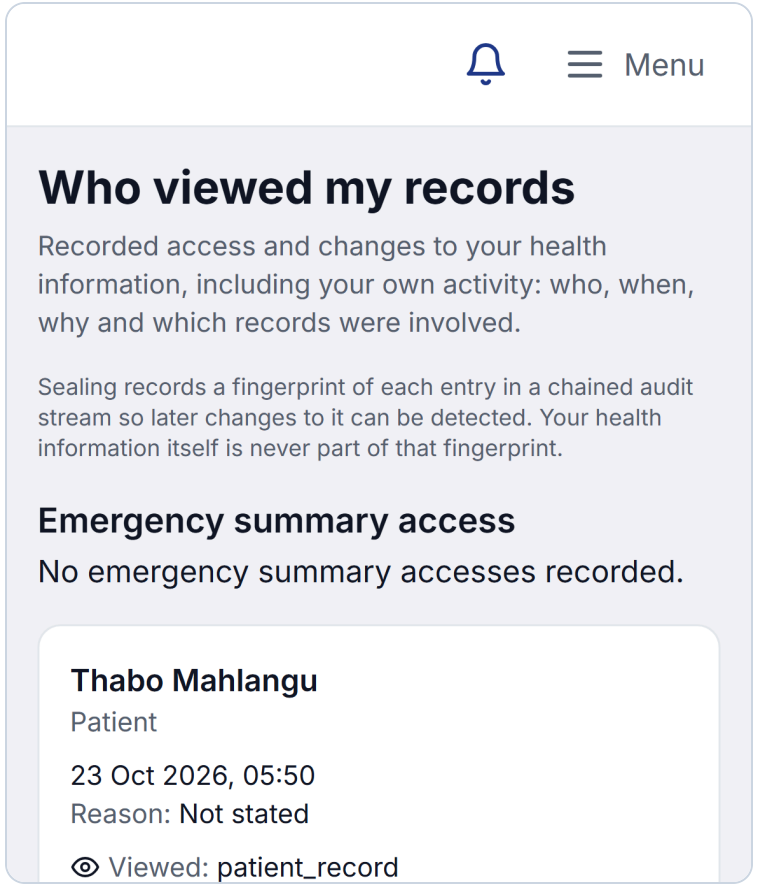



Figure 4. Emergency (break-glass) access. Only step 6, the durable audit row, may refuse a read; sealing never does.

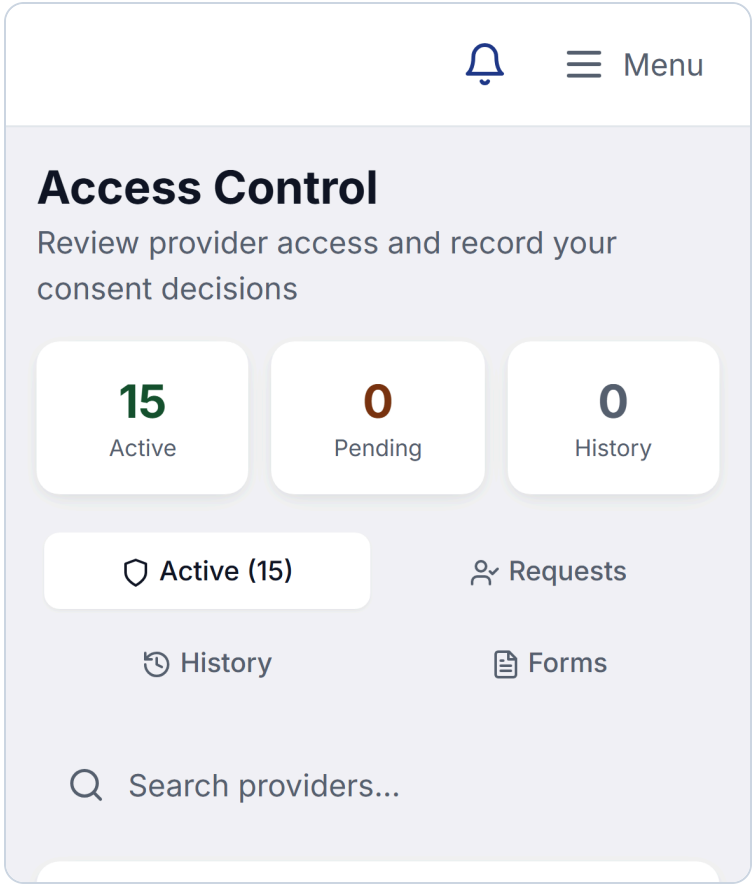
# Patients see who opened their record




Emergency card: the QR a responder scans



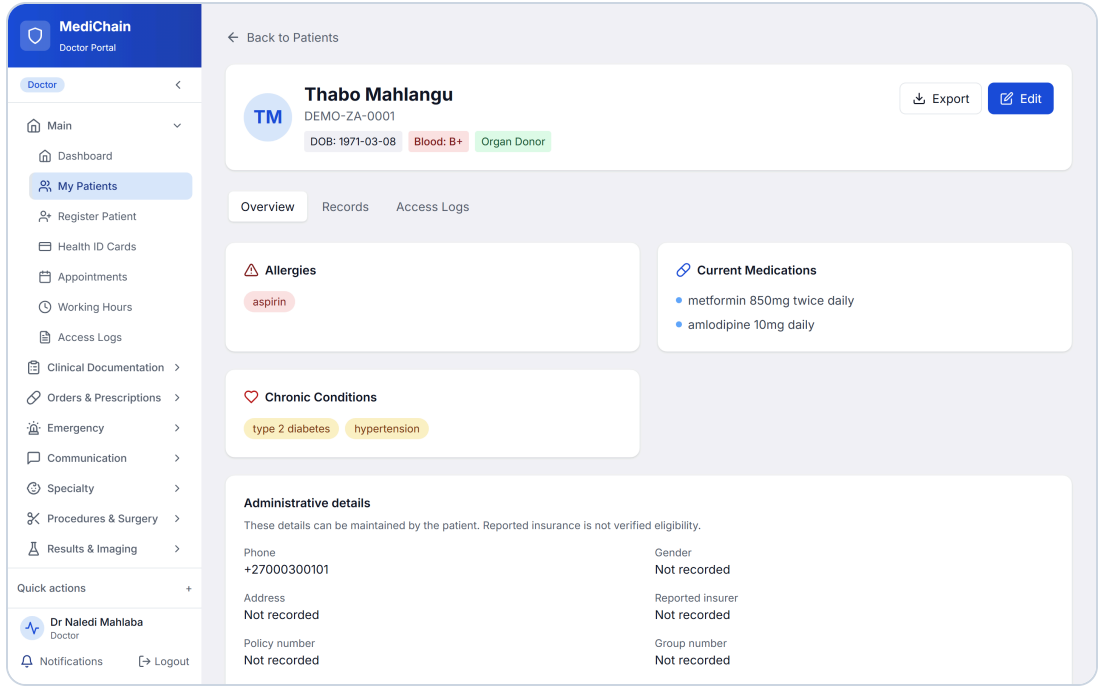

Access history: who, when, why




Access control: approve and withdraw

# Seven roles, one care loop

- **Patient:** records, medicines, consent, access history, messages
- **Doctor:** charts, SOAP notes from templates, orders, prescriptions
- **Nurse:** vitals, care plans, medication administration, handover
- **Lab technician:** specimens, results, critical values
- **Pharmacist:** prescription checks and dispensing
- **Paramedic:** emergency access and ambulance handover
- **Administrator:** staff and approved devices; no clinical access



A patient chart in the staff portal



All data shown is fictional

# Architecture: hosted in South Africa

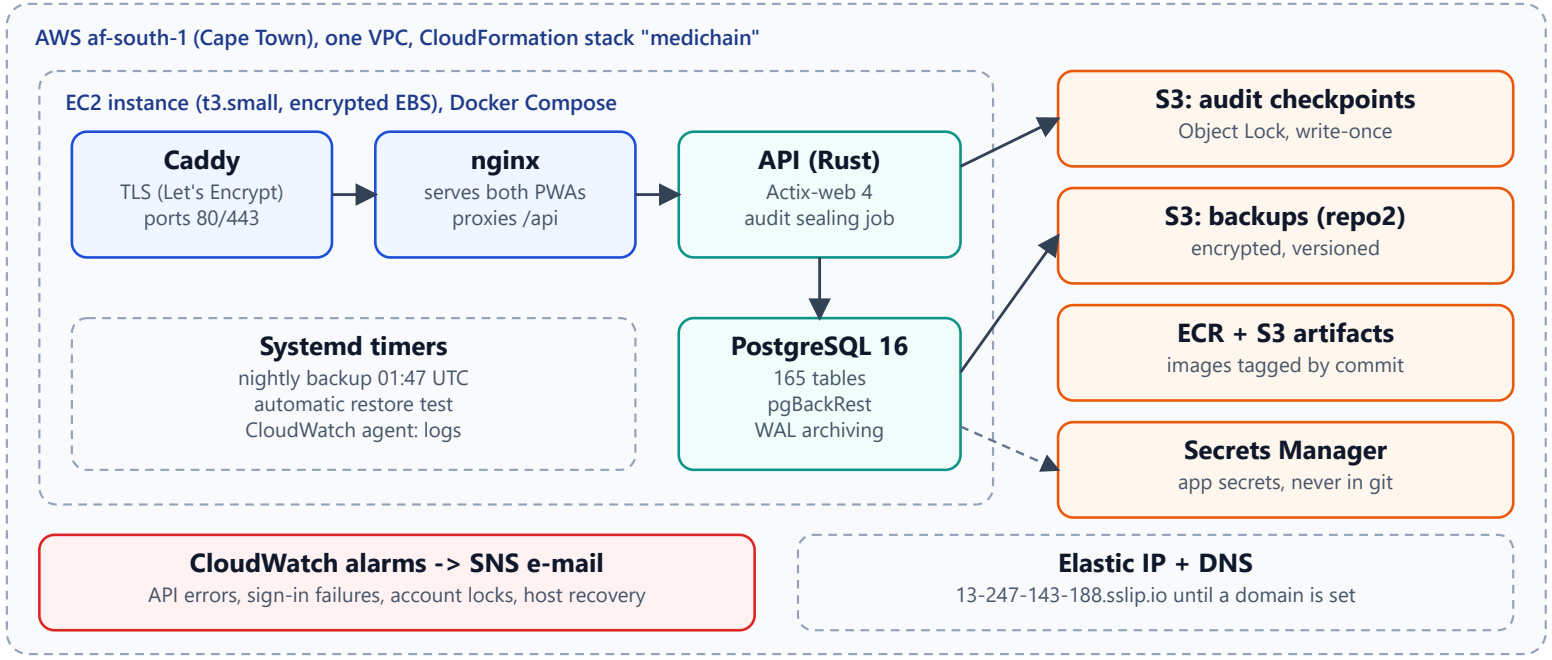



Figure 2. Deployment: everything runs in South Africa; data at rest is encrypted.

- Rust API (Actix-web), React web apps, PostgreSQL 16 with 165 tables
- Every request passes nine checks, then role and care-relationship rules
- Passwords never leave the browser; Argon2id verifiers
- Live updates by server-sent events, only to the right accounts
- One-script deploys; nightly backups with an automatic restore test

# Trust is designed in, for POPIA

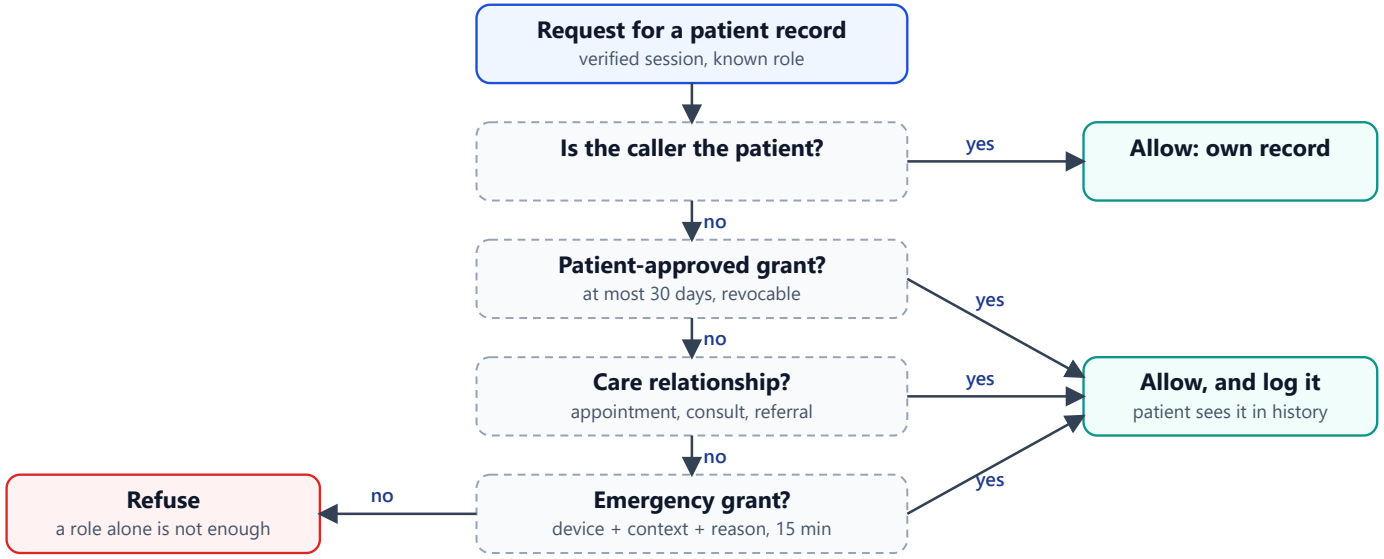



Figure 6. How the server decides whether to open a record. Administrators manage accounts and devices, never clinical records.

- Mapped to all eight POPIA conditions for lawful processing
- A role alone opens nothing; the patient sees every read
- Emergency access: enrolled device, reason, 15 minutes, one patient
- Audit batches signed and stored write-once (S3 Object Lock)
- Encrypted documents, disks and backups; data stays in Cape Town
- Fictional data only until POPIA authorisation and legal sign-off

# Built, live and tested

**1,045**
server tests passing

**276**
step journeys: every role fills in its forms; a colleague reads them back

**116**
screens across the patient app and the staff portal

- Live on AWS Cape Town with automatic HTTPS and alerts on errors and failed sign-ins
- 855 staff-portal and 280 patient-app tests; 35 automated code checks in CI
- Clinical scores (qSOFA, SOFA, TIMI, START triage and more) are explicit server rules, not predictions



All data shown is fictional

# Market and business

## Customers and model

- First private EMS operators, then clinics and GP groups, then hospital groups
- Subscription per facility or vehicle; always free for patients
- Under USD 60 a month to run today; cost grows with facilities, not patients
- Pilot (3 months), published proof (6 months), then Gauteng and the Western Cape

## Where it fits

| Today | MediChain adds |
|---|---|
| Paper folders | Records at the point of care; access recorded |
| OpenMRS, Bahmni | Patient-held summary and patient-visible access |
| iER | Clinician-maintained data on approved devices |
| Vula | Complementary: the patient's own record |

Comparisons describe focus, not shortcomings.

# The ask

A clinic, hospital or EMS partner for a supervised pilot,
and introductions to funders backing South African health technology.

## Next 3 months

Pilot partner, Information Officer, POPIA prior authorisation, legal sign-off, custom domain

## Next 6 months

Android app for phone-to-phone NFC and offline use; more languages; published pilot results

## Team

Keorapetswe Kgoatlha, founder-developer. First hires: a clinical lead and a second engineer

Try it: https://13-247-143-188.sslip.io/patient/ and https://13-247-143-188.sslip.io/doctor/ (logins in the judge guide)

kkgawatlh9@gmail.com · Lukau Invasion (Pty) Ltd · 2026/600804/07